

Terala Samiksha Gupta

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Summary

AI/ML Engineer with hands-on experience building production-scale deep learning systems across NLP, reinforcement learning, and transformer architectures. Skilled in end-to-end ML lifecycle , from model architecture, LLM fine-tuning, and embedding engineering to FastAPI deployment on AWS , with active exposure to agentic AI and UAV-integrated intelligence systems.

Education

Keshav Memorial Group of Institutions	Bachelor of Technology • 8.12 / 10
Computer Science & Engineering (Data Science)	Hyderabad , India • 2023 - 2027

Skills

Languages: Python, Java, JavaScript, SQL, C

AI & ML : PyTorch, TensorFlow, Scikit-learn, Transformers, NLP, Deep Learning, MoE, RL, LLM Fine-Tuning, Transfer Learning, Hyperparameter Tuning, AMP, Focal Loss, EMA

Data & Embeddings: NumPy, Pandas, LegalBERT, ChemBERTa, BPE Tokenization, Matplotlib

Infrastructure: FastAPI, Docker, AWS (EC2, S3), GitHub Actions, HuggingFace Hub

Full Stack & DBs: MERN, REST APIs, JWT, RBAC, MongoDB, MySQL

Experience

OSCOWL AI	Remote
AI Engineer Intern	May 2026 – Present
• Built a PPO-based RL system for adaptive RF null steering on a 16-element phased array, achieving null depths below -30 dB against dynamic jamming scenarios in simulation. • Contributed to an autonomous PX4-based UAV flight control platform with AI-driven control and telemetry, proposed for ISRO collaboration. • Built Python automation pipelines for model training, evaluation, and experiment tracking at scale.	
Keshav Memorial Group of Institutions	Hyderabad , India
Full Stack Developer	Jul 2025 – Jan 2026
• Designed, built, and deployed two production MERN stack applications serving 2,000+ students, implementing secure authentication, role-based access control, and RESTful API architecture. • Optimized MySQL and MongoDB database operations across 20,000+ records, improving query performance and data reliability; collaborated with cross-functional teams in an agile development environment.	

Projects

<u>Court-MOE: Legal Document Intelligence System</u>	Jul 2025 – Present
• Built a Mixture-of-Experts NLP system trained on 970K legal documents with 6.3GB LegalBERT embeddings, achieving 61.4% routing accuracy on NVIDIA A100 infrastructure. • Engineered production-scale training pipeline with AMP, focal loss, EMA, and custom embedding generation for stability and performance.	
<u>TransPolymer: Transformer Prediction System</u>	Feb 2025 – Jun 2025
• Designed a custom Transformer with BPE tokenizer and ChemBERTa embeddings with six parallel property prediction heads for multi-target polymer regression. • Deployed via FastAPI on AWS (EC2 + S3), automating the full ML lifecycle from training to production serving.	

Achievements & Certifications

- 500+ DSA problems solved across LeetCode and competitive programming platforms.

- Google Generative AI Learning Path - 11 Skill Badges (LLMs, Prompt Engineering, Emerging AI).
- Head of Event Management Committee, KMCE - led cross-functional teams for large-scale campus events.